

Antibiotic Resistance in *E. coli*

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Date: October 15, 2026

Introduction

Antibiotics are used to treat infections caused by bacteria. Antimicrobial resistance (AMR) occurs when microorganisms become resistant to antimicrobial drugs, making infections more difficult to treat. This is a growing concern because AMR limits and reduces the effectiveness of treatments for infections. In 2019, more than 1.2 million of deaths globally were directly linked to AMR [3]. This report aims to investigate which antibiotics have the highest resistance rates in *Escherichia coli (E. coli)*.

Data sample

The dataset [1] used for the study was obtained from the Bacterial and Viral Bioinformatics Resource Center (BV-BRC). The dataset contains over 195,000 observations of *E. coli* isolates and their antimicrobial susceptibility. The dataset includes genome IDs, strain names, antibiotics, MIC or zone diameter measurements, testing methods, and five resistant phenotype classifications: Resistant, Non-susceptible, Intermediate, Susceptible, and Susceptible-dose dependent.

For this analysis, the main variables of interest were "Antibiotic" and "Resistant Phenotype". The dataset contained 92 different antibiotics.

Data Cleaning

The dataset was cleaned and preprocessed to improve data consistency and ensure reliability for the analysis. During initial exploration of the dataset, it was found that there were missing data for phenotype classifications. In order to perform a complete analysis without discrepancies, observations with missing data were removed. Subsequently, the dataset contained 95,178 usable observations and 77 different antibiotics.

For the final resistance-rate comparison, only antibiotics with at least 100 observations were included. This threshold was used to reduce the influence of resistance rates calculated from very small sample sizes. After applying this criterion, 46 different antibiotics remained for the final resistance-rate analysis.

Data Analysis

The resistance rate and standard error of the resistance rate for each antibiotic were calculated using the following formulas. The antibiotics were then ranked by their resistance rate in descending order and the top 10 antibiotics were selected for the main comparison.

$$\text{Resistance Rate (\%)} = \frac{\text{Number of resistant observations}}{\text{Total classified observations}} \times 100$$

$$SE(\%) = \sqrt{\frac{p(1-p)}{n}} \times 100$$

where p is the resistance rate expressed as a proportion and n is the number of observations for the antibiotic.

The phenotype distribution was also calculated for each antibiotic to further examine the distribution of each resistance phenotype classification. Finally, the relationship between sample size and resistance rate was analyzed using the Pearson correlation coefficient. This was used to determine the relationship between an antibiotic's sample size and its resistance rate. The coefficient ranges from -1 to 1, with values close to 0 indicating a weak linear relationship [2].

Results and discussion

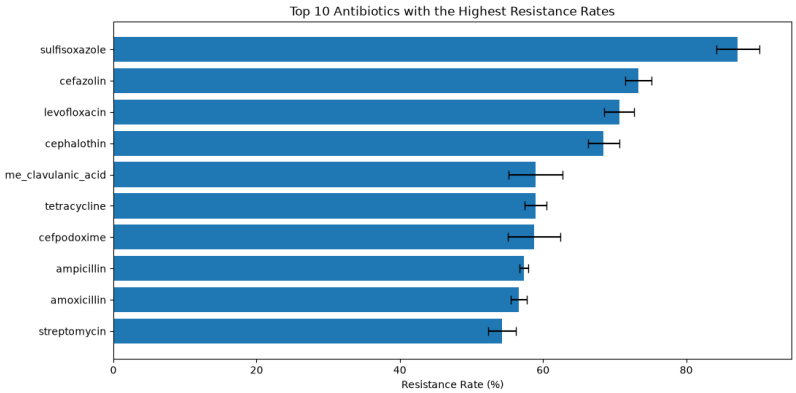


Figure 1: Top 10 antibiotics with the highest resistance rates in *E. coli*. Error bars represent the standard error of the resistance rates.

Figure 1 highlights the top ten antibiotics with the highest resistance rates and their respective standard errors. The antibiotic with the highest resistance rate is sulfisoxazole at 87.20 ± 2.99%. The phenotype distributions were analyzed to determine the percentage of classifications that were not resistant. For sulfisoxazole, 11.20% were susceptible and 1.60% were nonsusceptible. Most of the other antibiotics in the top ten had a larger proportion (approximately 20-40%) of observations classified as susceptible.

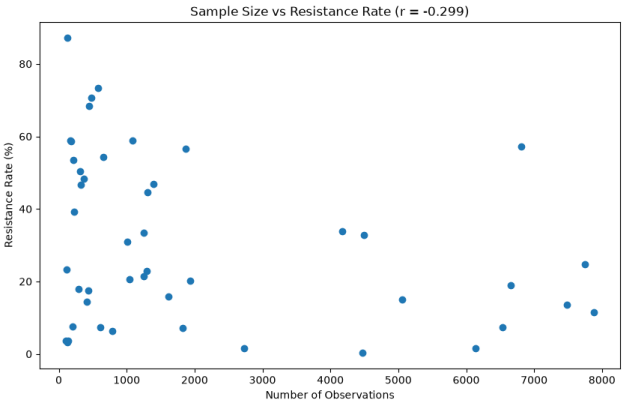


Figure 2: Relationship between sample size and resistance rate for the antibiotics included in the final analysis.

Figure 2 shows the relationship between sample size and resistance rate. The correlation coefficient was -0.299, a weak negative correlation. For this dataset, antibiotics with larger sample sizes tended to have lower resistance rates. However, the relationship was weak, indicating that sample size alone does not explain the variation in resistance rates.

A limitation of this analysis is that many of the observations were excluded because they had missing data and small sample sizes were excluded in the final analysis. Although these criteria made the analysis more refined, it also meant that some antibiotics were not included. Therefore, the results are only representation of the antibiotics included in this dataset.

Overall, these analyses addressed the research question of which antibiotic showed the highest resistance in *E. coli*. Based on this dataset, sulfisoxazole had the highest resistance rate at 87.20%.

References

[1] Maciel, V. (2025). *E. coli Resistance Dataset*. Kaggle. Available at: https://www.kaggle.com/datasets/valeriamaciel/e-coli-resistance-dataset.

[2] Schober, P., Boer, C., & Schwarte, L. A. (2018). *Correlation coefficients: Appropriate use and interpretation*. Anesthesia & Analgesia, 126(5), 1763–1768. Available at: https://doi.org/10.1213/ANE.0000000000002864.

[3] Tang, K. W. K., Millar, B. C., & Moore, J. E. (2023). *Antimicrobial resistance (AMR)*. British Journal of Biomedical Science, 80(11387), 11387. Available at: https://doi.org/10.3389/bjbs.2023.11387.